



Plants for amelioration of subsoil constraints and hydrological control: the primer-plant concept

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Abstract

In this review, we propose the use of suitable plant species, termed *primer-plants*, for the primary purpose of preparing soil conditions for the benefit of following crops. Such plants may be used in the temperate agricultural belts of southern Australia, where dryland salinity is a major environmental and agricultural problem that threatens the viability of many crop production enterprises. It is recognised that growing plants that have deeper roots and use more water than the current shallow-rooted annual crops provide a long-term solution for managing the dryland salinity problem. Increased plant water-use is expected to mitigate the rising watertable that transfers salt to the root-zone of crop plants. On medium to heavy textured soils, common in this region, impermeability of the subsoil to roots and water movement is another major impediment to high water-use and productivity by plants, which may lead to other adverse hydrological events in the soil such as water-logging and excessive run-off. Plants that possess the ability to penetrate the dense subsoil and make it porous, in addition to having the capacity for using soil water at high rates, should be effective in combating dryland salinity. These plants normally should have thick roots that grow deep in the soil and are able to modify or withstand the adverse chemistry of the often-saturated subsoil, so that upon the death and decay of their roots, channels or *biopores* are created. These biopores have greater vertical and lateral continuity and last longer than porosity created through mechanical tillage. In this paper, we argue that potential exists for inclusion of short to medium-term phases of primer-plants in farming systems as a mimic of pre-existing perennial vegetation. We propose that ideal plants for combating dryland salinity should have high water-use and capacity to also improve soil structure and, possibly, nutrition. Examples are presented of soil amelioration that generally supports the viability of primer-plant concept, including the limited work undertaken in south-eastern Australia. We identified key knowledge-gaps, such as lack of well-defined agronomic packages for growing short-phases of Australian native species in mixtures, and our limited understanding of their root dynamics, which need to be addressed before effective implementation of the primer-plant concept.

Introduction

Management of land degradation such as soil structural decline, adverse soil chemistry and erosion is a major issue for sustainable agricultural practices in many parts of the world. In Australia, land degradation

caused by dryland salinity is a monumental challenge to agricultural industry, and in some cases the continuity of whole agricultural enterprises is being jeopardised. The fundamental cause of degradation is generally associated with the replacement of original vegetation with other species that are of exotic provenance, in addition to farming practices that continuously mine the soil. In a major review of dryland

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salinity, Hatton et al. (2003) argued that a mere lowering of the watertable may not restore the health of the landscape, since significant changes have occurred in the soil structure and chemistry. It is imperative that strategies designed to restore water-balance should also involve restoration of soil structure. Soil and the vegetation it supports co-evolved over time, such as the red earths of the tropics and the brown earth of temperate regions that developed under natural woodlands and forests (Young, 1986). A long-term solution to dryland salinity and soil degradation may, therefore, involve the use of plant species that possess the relevant features of the original vegetation, if they can be successfully established and managed. These plant species should be capable of drilling through difficult subsoils, and changing their structure, thereby providing an environment that is conducive to the roots of following generations of plants gaining access to increased water and nutrient supply. Plants modify soil structure mainly when their roots grow into the dense layers, then later die and decay leaving behind pores or channels that are termed *biopores*. The Macquarie English dictionary defines a 'primer' as anything that 'prepares a material for a specific purpose or operation'. Thus 'primer-plants' are plants grown for the specific purpose of soil management. We use this term here specifically to refer to cropping systems where short-phases of appropriate plant species are grown to condition the soil for following crops of high economic value. It thus differs from the more generic term of phyto-remediation, which refers to the use of a range of plants orders (e.g., algae, moss, herbs, shrubs) within a defined time frame to ameliorate a myriad of environmental problems (Qadir and Oster, 2002). It is only in recent times that plants with mechanically resilient root systems have been intentionally used as biological drills to overcome high soil strength (Elkins and Van Sickle, 1984; Elkins et al., 1977), and a wide application of plants for soil management is still limited. *Primer-plants* need not necessarily have direct economic benefit, but should be effective in conditioning the soil for crops of economic value and in conserving environmental resources.

Numerous studies have demonstrated the use of plants in addressing local and/or regional soil management problems around the world. In southern Nigeria, where a dense subsurface gravel layer impedes water movement and root growth, Hulugalle and Lal (1986) found that pigeon pea (*Cajanus cajan*) improved the structure of the layer, which enabled the following maize (*Zea mays*) to increase its root length densit-

ies and grain yield by almost 31%. Jama et al. (1998) in eastern Africa, demonstrated the economic benefit of including leguminous trees of sesbania (*Sesbania sesban*) in rotation with maize (*Zea mays*) rather than the traditional unmanaged fallow. In North America, Elkins and van Sickle (1984) increased the number of biopores (>1 mm) by growing bahia grass (*Paspalum notatum*) for 5 years on soils with plow pans caused by farm traffic. This enabled the roots of the following cotton to grow beyond 2 m depths and to increase boll yields by 300%, compared to soil under continuous-cotton where there was an 87% reduction in the number of biopores and roots were restricted to less than 0.3 m depth. The use of eastern gamagrass (*Tripsacum dactyloides*) to manage claypan soils of the Midwestern prairies of the United States is a classical example of soil structural improvement through biological drilling (Comis, 1997). This grass thrives on heavy and impermeable clayey subsoils and is being intensively studied for managing structurally difficult soils in other parts of the country (Clark et al., 1998; Comis, 1997). The soil primed with the grass increased root growth by the neighbouring soybeans (*Glycine max*) beyond the normal depth of 0.60 m.

Qadir and Oster (2002) provided an excellent example of how lucerne (*Medicago sativa*), sesbania (*Sesbania aculeata*) and Kallar grass (*Letochloa fusca*) were used to manage saline-sodic soil in the Indian subcontinent. For instance, Ilyas et al. (1993) reported a doubling of saturated hydraulic conductivity of the top profile of the soil a year after planting either lucerne or sesbania compared to mulched soil. An early development of primer-plant strategies could mitigate current and potential problems associated with land degradation in China, where large scale land clearance is ongoing and major problems of declining soil organic matter, erosion, and desertification have intensified (Ye et al., 2002). In such places, strategic use of woody perennials in cropping systems from the onset can maintain soil structure more effectively than continuous annual cropping (Ghuman et al., 1991).

There have been several isolated studies on the use of plants as tillage tools in Australia (e.g., Aylmore, 1995; Cornish, 1993; Cresswell and Kirkegaard, 1995; Lorimer and Douglas, 2001; Materechera et al., 1993; Pillai and McGarry, 1999; Yunusa et al., 2002b). Some of these previous works are considered at length in this review. Only few of these, however, evaluated both the priming and cropping phases, and the main soil processes, such as water and nutrient use, during and after the priming phases. This situation led to a con-

Table 1. Subsoil structural characteristics for optimum root growth and function compiled by Crockroft and Olsson (1997) and the values commonly observed under crops on the Australian duplex soils

Structural attribute	Optimum values	Observed range	References for the observed values
Mechanical resistance (MPa)	<0.5	0.1–3.5	Hamblin (1984); McKenzie and McBratney (2001)
Air-filled porosity (%) ^a	>15%	2.5–8.7	Yunusa et al. (2002b)
Volumetric water content at field capacity (m ³ m ⁻³)	>20%	14–40	Hamblin (1987); Yunusa et al. (2002b)

^aDetermined at –10 kPa water potential.

clusion by Aylmore (1995) that the role of plant roots in maintaining soil structure is underestimated, while the potential ‘biological drilling’ ability of plants for managing soil structure ‘is virtually unexplored’. In this review, we used the Australian agricultural landscape to (1) examine the concept of growing adaptive plant species, including native woody species and deep rooted herbaceous species, primarily for their biological tillage capability, (2) discuss criteria for effective primer-plants, and (3) identify knowledge-gaps to be addressed before adoption of the concept can be achieved.

Dryland salinity and land degradation in Australia

About 2.5 million ha of agricultural land are affected by dryland salinity in agricultural lands of southern Australia (Barr and Cary, 1992). This has developed mainly due to mobilisation of salt from deep within the profile by excess soil water draining into the watertable, a process that is still continuing, especially in the 400–600-mm rainfall zones. Salinity problems became more widespread when the native grassland and woodland vegetation, dominated by perennial species, were replaced with mostly annual species of crops and pastures, which are shallow-rooted and use less of the rainwater stored in the soil. Water then continued to be added to the underlying water table, which then rose and brought salts to the shallow layers and surface of the soil. Between 12 and 15 billion trees, whose roots could reach depths in excess of 20 m, were believed to have been removed beginning in the 19th Century from the Murray-Darling basin alone (Walker et al., 1993). Williams et al. (2001) argued, however, that the watertable is recharged mostly by run-off, and preferential flows through *holes* or *holeyness* within landscapes, during or following rainfall events, rather than by drainage from a saturated soil

matrix beneath crops and pastures. They associated increased run-off with changes in soil characteristics that have reduced infiltration, and contended that rather than being saturated, the soil matrix in the root zone might in fact be getting drier over time. This hypothesis is not strictly supported by several studies that linked deep drainage to saturated soil profiles in cropping lands (Dunin et al., 1999, 2001; Passioura and Ridley, 1998; Ridley et al., 2001). The ability of regenerating and undisturbed natural wooded vegetation to use almost all the water from rainfall in any given season is recognised (Nulsen et al., 1986; Schofield and Bari, 1991). These vegetation systems that are dominated with perennial species are unlike annual pastures and crops beneath which up to 43% of water from annual rainfall could be drained beneath the rooting depth (Carbon et al., 1982; Passioura and Ridley, 1998). It is, therefore, unlikely that restoration of the hydrological balance, by re-introducing perennial vegetation, alone would be enough to ensure sustainable land-use enterprises, without amelioration of the physical and bio-geochemical characteristics of the soil (Hatton et al., 2003; Moran, 1998; Russell and Isbell, 1986; Williams et al., 2001).

Many soil types in southern cropping zones are characterised by strong contrasts in texture, hydrology, and inherent conditions for plant growth between A- and B-horizons (Northcote, 1960). They show huge diversity in their parent materials, mode of development, and other properties. These soils, generally termed *duplex soils*, constitute about 20% of the soils along the east and southwest coastlines, and eastern hinterlands, and are mostly sodic and/or saline (Chittleborough, 1992). A general feature of these soils is the impermeable clayey B-horizon, in which the bulk density increases while hydraulic conductivity declines with depth (Blaikie and Mason, 1993; Olsson et al., 1995). Therefore, waterlogging is common on these soils, especially in the medium to high rain-

Table 2. The presence of growing roots found at different levels of soil structural constraints (density or strength)

Plant species	Source	
	Bulk density (Mg m^{-3})	
Ryegrass (<i>Lolium perenne</i>)	1.3	Cornish (1993)
Wheat (<i>Triticum aestivum</i>)	1.6–1.7	Yunusa et al. (2002b)
Mixed woody species ^b	1.8	Yunusa et al. (2002b)
	Soil strength (MPa) ^a	
Corn (<i>Zea mays</i>)	3	Laboski et al. (1998)
Eucalypts (<i>Eucalyptus alba</i>)	>6	Laclau et al. (2001)
Radiata pine (<i>Pinus radiata</i>)	<3	Sands et al. (1979)

^aValues for the dense layers.

^bConsisted of *Eucalyptus* spp., *Acacia* spp and *Casuarina* spp.

Table 3. Broad classification of pores, their function and origin

Pore diameter (μm)	Function ^a	Equivalent water potential (m) ^b	Origin ^c
>5000	Bypass flow	<0.006	Roots and macrofauna, shrinkage cracks
500–5000	Niches for macrofauna	0.06–0.006	Dicot roots, nodal and seminal roots of cereals, worms, ants, termites
	Rapid water and air transmission		
	Niches for microbes and arthropods	0.10–0.06	Seminal roots of cereals
300–500	Water conduction and aeration		
30–300	Water conduction	1.0–0.1	First and second order lateral roots of cereals
0.20–30	Storage of plant available water	150–1.0	Root hairs, fungal hyphae, bacteria, enchytraeids
<0.20	Residual pores, storage of unavailable water	>150	Fungal hyphae, bacteria

^aAfter Greenland (1981) and Hamblin (1987).

^bCalculated with Equation (1). ^cFrom Hamblin (1984), Shipitalo et al. (1990) and Van Vliet et al. (1998).

fall areas, which cover about 1 M ha in southern New South Wales and Victoria alone. Their sodicity is defined by an exchangeable sodium percentage (ESP) in excess of 6 (Rengasamy and Olsson, 1991), which further predisposes them to structural degradation, dispersion, and poor aeration.

Peck and Hurle (1973) argued that dryland salinity could be prevented, and salinised land reclaimed, by increasing water-use by plants, which can be achieved by growing species that use more water than currently grown crops. To successively implement this suggestion, the soil conditions need to be conducive for the 'new' high water-use plants. Cockroft and Olsson (1997) compiled attributes of soil quality for optimum root growth and function of duplex soils in terms of porosity, mechanical resistance, aeration, soil water

and temperature (Table 1). These ideal structural attributes, however, are rarely attained in practice on agricultural soils, but are found more often in forest soils where macroporosity of up to 28% have been reported in a silty clay loam (Incerti et al., 1987). Air-filled porosity of 2.5% was observed under annual crops (Table 1), compared with 8.7% for soil that supported a 6-year phase of woody species, which suggested that a woody primer-plant has the potential to open up dense soils. Deep-tillage and other mechanical means of disturbance prepare only a limited soil volume for root exploration (Olsson et al. 2002; Schmidt and Belford, 1994; Taylor and Olsson, 1987). These provide only temporary amelioration and cause soil structural degradation in the long-term (Hamblin, 1984; Jayawardene and Chan, 1994; Kirchhof et al.,

2000; Lal, 1989; Lorimer and Douglas, 1995). Vehicle traffic and farm equipment also destroy the integrity of macropores in the soil resulting in reductions in saturated hydraulic conductivity (K_{sat}) by up to 90% (Rab, 1994). The ability of plant roots to improve soil structure has long been recognised in the literature (e.g., Barley 1953; Clarke et al., 1967; Oades, 1993). In Table 2, examples are given of adverse soil structural conditions, where certain species were able to maintain growing roots. For instance, Laclau et al. (2001) found 25% of all fine roots of the eucalypts in the top 2 m of the soil where the soil strength exceeded 6 MPa; whereas roots of wheat and ryegrass were found in zones of lower densities than for woody species (Table 2). These data suggested that certain crop roots can withstand soils of high density, and so have the potential to create biopores within the subsoil.

Excellent literature reviews have been undertaken previously on the interactions between soil structure and soil water dynamics, tillage and soil biota (e.g., Kirby and Blunden, 1991; Letey, 1991; Oades, 1993; Topp et al., 1997), and plant root growth and water uptake (Hamblin, 1984). Creswell and Kirkegaard (1995) presented a brief theoretical discussion and experimental evidence for the use of roots as biological drills for ameliorating impermeable subsoils. In the following sections, we briefly discuss how replacement of native vegetation may have caused structural degradation of the subsoil. We provide examples of the primer-plant concept, including recent studies in south-eastern Australia, identify desirable characteristics of ideal primer-plants, define limitations of the concept, and propose a practical framework for incorporating phases of perennial species into current cropping systems.

Influence of vegetation cover on soil structure

Soil structure is the mutual arrangement, orientation, and organisation of the particles and pores in the soil (Hillel, 1971; Oades, 1993). The configuration of the particles and other biophysical activities result in the formation of irregularly shaped pore spaces that are highly interconnected and are important in the biological, hydrological and chemical processes in the soil. Ringrose-Voase (1987) identified two major types of pores as textural/structural and non-structural, each with subtypes.

Structural or textural pores are formed by the interaction of the texture, clay mineralogy and organic

matter content of the soil, and from the stresses caused by wetting and drying cycles (fissures or residual pores) (Ringrose-Voase, 1991). These pores are generally small ($<100 \mu\text{m}$ diameter) and termed micropores, thus, they determine structural units of the soil such as peds and aggregates and, to some extent, their arrangement. Non-structural pores are mainly macropores ($>100 \mu\text{m}$ diameter) created mostly through drilling by flora and fauna, i.e., biopores are macropores (Table 3). In terms of function, biopores in the range of $30\text{--}5000 \mu\text{m}$ ($<1 \text{ m}$ water potential), created by the thick roots of dicots, are sufficiently large to provide channels for new root growth and optimum water and air conduction (Table 3), but without excessive preferential flow. While structural pores are generally stable and able to withstand physical disturbance of the soil, biopores are easily disrupted by cultivation, and blocked by fine particles displaced by roots and by long periods of saturation (Richard et al., 2001). Luxmoore (1981) identified three broad classes of pores as micropores ($<10 \mu\text{m}$), mesopores ($10\text{--}100 \mu\text{m}$) and macropores ($>1000 \mu\text{m}$). Others considered any pore $>30 \mu\text{m}$, i.e., pores larger than those that hold water at field capacity (-10 kPa) as macropores (e.g., Creswell and Kirkegaard, 1995), which is the classification we have adopted in this paper. All biopores are generally too large to hold water by capillarity, but enhance water and air movement, and provide channels for the succeeding generations of roots (Bouma, 1981; Passioura, 2000). The function of pores in terms of distribution and storage of water is determined by their size (Hillel, 1971):

$$d = 0.03/h \quad (1)$$

in which d is diameter of the pore (mm) and h is the water potential (m). Table 3 also shows that large pores essential for mitigating water-logging are formed by roots and macroflora. Collis-George (1991) emphasised the importance of structural pores such as biopores, which are not entirely closed when the soil is wet, in draining clay and silt horizons, common in duplex soils. Thus, it is important for these biopores to remain stable irrespective of soil moisture contents.

Roots are essential for maintaining soil structure, and absence of roots at the lower depths of the soil profile under shallow rooting crops may stop or slow the processes of macropore formation. Plants improve soil porosity by either creating biopores or structural cracks (Czanes et al., 2000; Hamblin, 1984; Oades, 1993; Pillai and McGarry, 1999). Thus, forests

and permanent pastures with deep roots that experience successive wet-dry cycles over the years, often have good soil structure (Lorimer and Douglas, 1995; Sollins and Radunavich, 1988). In addition to pore creation, a permanent vegetation cover provides ancillary benefits, such as surface litter, for maintaining soil structure. A surface layer of litter stabilises soil moisture and temperature regimes, and increases organic matter to support soil-dwelling arthropods, and produces organic compounds that bind soil particles and buffer soil chemistry. Messing et al. (1997) rehabilitated degraded agricultural land by growing a mixture of deep-rooted aspen (*Populus deltoides*) and silver birch (*Betula pendula*), and reported that the trees were far more effective in increasing macroporosity and saturated hydraulic conductivity (K_{sat}), and in reducing bulk densities, than ley cropping systems. Planting of trees, may not by itself, necessarily enhance soil structure, but when strategically implemented in terms of species, density and duration of growth (age), desired changes in soil structure and hydrology can be achieved. This is important, since soil compaction may occur even in forests and other natural vegetation resulting from: (i) collapse of pores due to excessive drying (Allbrook, 1993; McGarry and Malafant, 1987), (ii) sealing of voids caused by rearrangement of particles by roots (Dexter, 1978), (iii) overburden pressure, (iv) low organic matter content (Sands et al., 1979), or (v) surface sealing by large raindrops (Hamblin, 1985). These processes explain the low hydraulic conductivity and poor porosity that have been reported for soils under forests and native vegetation in some instances (Kirchhof et al., 2000; Sands et al., 1979; Talsma and Hallam, 1980). These undesirable consequences for soil structure can be avoided with the primer-plant concept, since the woody species are to be grown in short-phases (≤ 5 years).

Influence of vegetation cover on soil hydrology

Soil hydrology is a function of storage and movement of water within the profile and is dependent on the pore-size distribution. Continuous macropores, especially those $>30 \mu\text{m}$ (Table 3), ensure high K_{sat} (Blackwell et al., 1990; Kirchhof et al., 2000). There is a significant correlation between number of macropores ($\geq 2.0 \text{ mm}$) with either $K(\theta)$, at a -10 mm supply potential, or air-filled porosity (Yunusa et al., 2002b). Lorimer and Douglas (1995) reported that

$K(\theta)$ at -10 mm supply potential for the clayey subsoil of a red-brown earth declined from 2808 mm day^{-1} by between 80 and 91% under a native forest of eucalypts, after only six seasons of continuous annual cropping. By comparison, the decline in $K(\theta)$ was only 60% where the soil was left under native pastures (*Danthonia* spp.). This was consistent with reductions in hydraulic conductivity of between 50 and 91% when native vegetation consisting of grasses and woody shrubs was replaced with grazed pastures on vertisols (Kirchhof et al., 2000). Thus, the primer-plant concept will be most effective in combating soil salinisation when planted in the recharge zones of the landscape, most of which is still productive. In such places, the primer-plants should ameliorate a range of adverse subsoil conditions including dewatering.

The fundamental principle of the primer-plant concept is the same as for the age-old principle of shifting agriculture, whereby natural ecosystems are used to repair disruption caused by farming. In the cropping environment of southern Australia, inclusion of a short-phase of woody species would gradually intercept and lower the watertable that built up during the cropping phase to depths well below those reached by roots of annual crops (Figure 1). During the priming phase, the thick roots of the woody species grow deep into the subsoil. Biopores are then created through the natural process of turnover of roots, and later through their decomposition after termination of the woody phase. Soil water storage capacity resulting from high rates of evapotranspiration by the woody species would reduce run-off and forestall drainage for several years in the following cropping phase as illustrated in Figure 1. Also, the absence of tillage during the priming phase enhances accumulation of soil organic matter and nutrients. The primer-plant concept, therefore, provides a viable new alternative to, or complements, aspects of conventional tillage techniques, which on their own are not sustainable on a large scale because of cost, ephemeral or superficial amelioration and potential for erosive losses of soil.

Instances of the primer-plant concept in Australia

On the hard-setting soils of Western Australia, Aylmore (1995) found longer-lasting amelioration of physical constraints and stabilisation of soil structure with plant roots, rather than with application of soil amendments, such as straw or gypsum. When wheat was grown on soil previously supporting thick-rooted

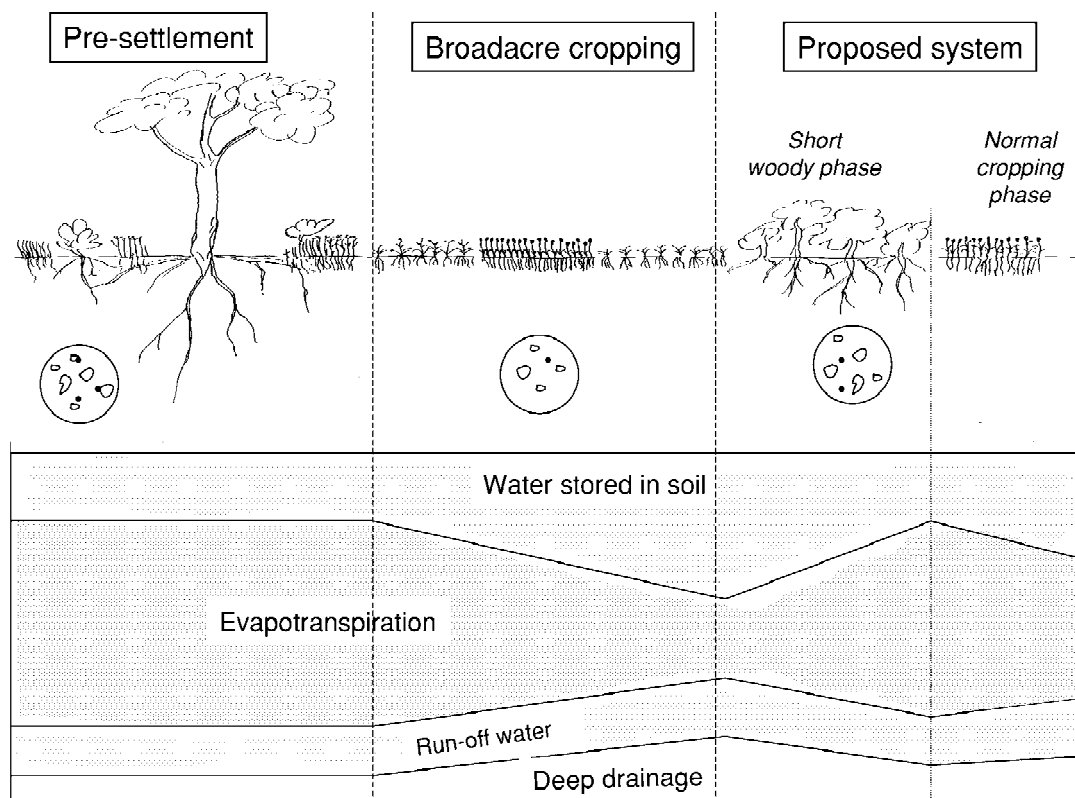


Figure 1. Representation of the landscape under the pre-existing vegetation before European settlement, its replacement with broadacre cropping, and the proposed method of alternating short-phases of woody species with annual crops, and the associated relative magnitudes of the components of soil water balance. Circles indicate relative pore size distribution in the subsoil.

legumes, its root growth was equal to or better than on soil treated mechanically or with soil amendments. Although pore size characterisation was not undertaken in that study, soil desegregation, as a measure of pore instability, was reduced by between 6 and 9%, while water soluble aggregation was increased by between 6 and 27%. Legumes with thick roots such as lupins (*Lupinus* spp.) were found to be more effective than non-legumes, while annual ryegrass (*Lolium rigidum*) with fine roots was least effective in improving the soil structure. In an irrigated trial on highly clayey vertisols (Typic Haplustert) in northern Australia, Pillai and McGarry (1999) found that increases in clod porosity and interconnected pore space under thick-rooted mung bean (*Vigna radiata*) and lab lab bean (*Lablab purpureus*) were twice that under sorghum (*Sorghum bicolor*) and wheat. It also took the beans only three wet–dry cycles to achieve soil improvement compared to six cycles needed by the cereals. This implied that in

dryland situations several wet and dry seasons would be required for plant roots to effectively modify soil structure, which probably explains in part the contrasting results obtained in southern Australia (Cresswell and Kirkegaard, 1995; Yunusa et al., 2002b).

Studies by Cresswell and Kirkegaard (1995) and Yunusa et al. (2002b) evaluated major physical attributes of the impermeable clayey subsoils under different vegetation systems in south-eastern Australia. Cresswell and Kirkegaard (1995) compared canola (*Brassica napus*) and wheat for their ability to create *biopores* on top of the B-horizon, but found no differences between the two treatments in their modifications of major structural and hydrological parameters, including number of pores (>2.0 mm), bulk density and $K(\theta)$ (Table 4). A 12.5% increase in the grain yield by the subsequent wheat crop in the canola–wheat rotation, relative to continuous wheat, was deemed to be associated with issues other than biological drilling.

Table 4. Comparison of soil structural variables on Vertic luvisol at Temora (36°08'S, 146°28'E) and on Haplic xerosol at Rutherglen (34°26'S, 147°32'E) under different land-use systems in south-eastern Australia

Soil structural variables ^a	Temora ^b		Rutherglen ^c	
	Continuous wheat	Wheat–canola	Continuous cropping	Woody species
Bulk density (Mg m ⁻³)	1.65	1.66	1.68	1.72
Pores >2.0 mm (number m ⁻²)	235	196	97	161
$K(\theta)$ (mm day ⁻¹)	110	139	128	310
Mean θ (m ³ m ⁻³) at planting of following crop	0.15	0.14	0.31	0.18
Grain yield for the following crop (g m ⁻²)	320.0	360.0	856.6	527.1

^aPore numbers and bulk density measured near top of B-horizon; hydraulic conductivity ($K(\theta)$) was for B-horizon; following crops were wheat at Temora and oat (*Avena sativa* L.) at Rutherglen.

^bCresswell and Kirkegaard (1995). ^cYunusa et al. (2002b), except volumetric soil water and grain yield from Mele and Yunusa (2001).

The authors acknowledged that the soil might be too dense for canola roots to penetrate, especially within the limited period of a growing season, i.e., the primer plant, which was canola in this case, did not experience several wet-dry cycles suggested by the study of Pillai and McGarry (1999). Cresswell and Kirkegaard (1995) concluded that deep-rooted perennial species such as lucerne (*Medicago sativa*) could be more effective. An annual growth cycle might, however, not be the only constraint on the ability of roots to penetrate a dense soil profile, since Materechera et al. (1991) identified increased thickness of roots as a major factor in this process.

Evidence of soil priming by plants was found in a later study by Yunusa et al. (2002b) on a soil similar to that of Cresswell and Kirkegaard (1995), but involving a 6-year phase of Australian native woody species (*Acacia* spp., but containing *Eucalyptus* spp. and *Casuarina cunninghamiana*). These species had all the features of effective soil primers; they were perennials, had thick roots that grow deeply, and native to the general geographical region. There were increases in the number of large pores (>2 mm) of 68% and in $K(\theta)$ of 100% in soil that previously supported the woody species, compared with those under annual crop rotations (Table 4). The woody species, however, dewatered the profile to the extent that the bulk density measured shortly after the removal was high, partly due to compaction during penetration of thick roots as discussed previously or temporary collapse of storage pores due to excessive drying. With a gradual decomposition of the thick roots and a rehydration of the soil, permeability improved when the measurements were repeated 22 months after the removal of woody species. At this time, the number of large

pores was almost doubled and preferential flow 50% larger, respectively, than found in the continuously cropped soil (Yunusa et al., 2002b). Improvements in the permeability of soil doubled root length density and increased root mass by 43% for wheat on the primed soil (Table 4), but failed to increase crop yield because of weed infestation and an initially drier soil profile in the primed soil (Mele and Yunusa, 2001). It was also found in this study that both $K(\theta)$ and air-filled porosity increased with number of biopores (Figure 2). The curvilinear response in air-filled porosity to number of pores was associated with a high proportion of superficial pores in the topsoil layers. The responses in Figure 2 were consistent with data presented in Table 3, which showed internal drainage to be dependent on the large pores. Hence, unlike canola in the study of Cresswell and Kirkegaard (1995), a combination of perenniality (i.e., several wet-dry cycles) and thick roots enabled the short-phase of native woody species to significantly increase number of biopores, and use almost all the available water in the soil profile.

Primer-plant induced changes in soil processes

Changes to soil structure through plant-induced increases in macroporosity have the potential to influence soil water dynamics, including timing and rate of evapotranspiration and nutrient uptake, aeration, and redistribution of particles and solutes (White, 1985). These factors would influence chemical and microbial activities in the rhizosphere and subsequently the growth and yield of the following crop.

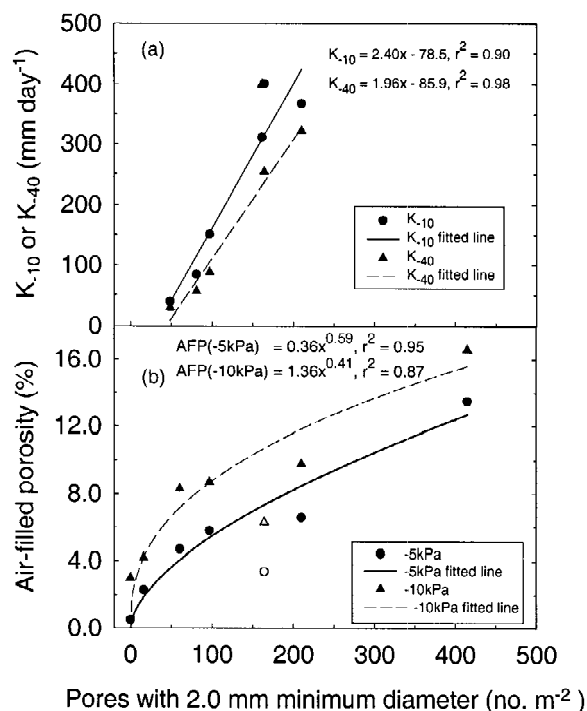


Figure 2. Relationships between number of pores having diameters ≥ 2.0 mm and either (a) hydraulic conductivity or (b) air-filled porosity 10 months after belt removal. In (a) there were no hydraulic conductivity data for the pasture treatment; open symbols in both graphs are outliers excluded from the regressions due to large variability (Source: Yunusa et al., 2002b).

Soil water dynamics

Perennial primer-plants change hydrological conditions by either one or more combinations of (a) dewatering during the priming phase to provide increased storage capacity for rainfall, (b) increasing K_{sat} and/or preferential flow, and (c) potential increase in water-use by the following crop through a greater exploration of the soil profile. Extended periods of seasonal growth by perennial species ensures continuous use of soil water, making them suited for drying the soil profile in high rainfall environments (Passioura and Ridley, 1998; Ridley et al., 2001; Ward et al., 2001). Ward et al. (2001) found that after 3 years of lucerne the first wheat crop used 35 mm more water than wheat grown after clover pasture, despite comparable winter re-wetting in the 0.45–1.2-m soil depth. Native woody perennials may however, have a greater potential for dewatering than lucerne. A comparison of mixtures of native woody species in the study of Yunusa et al. (2002a) and lucerne on a duplex soil with clayey subsoil, showed that the native woody species

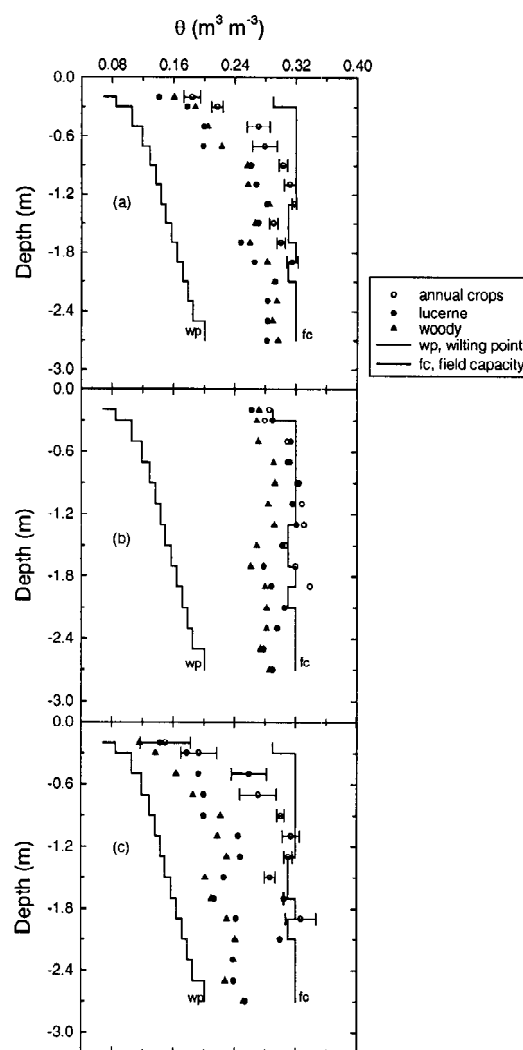


Figure 3. Distribution of water in the profiles of soil under either an annual crop, lucerne or native woody species on (a) 3 November 1995, (b) 8 October 1996, and (c) 5 July 1997. Both lucerne and woody species were sown in September 1993. Also shown is the water content at nominal permanent wilting point (wp) and at field capacity (fc). (Source: Yunusa et al., 2002a).

had a greater tendency than lucerne to dry the soil (Figure 3). It is noteworthy that despite the similarity in water content at individual depths for woody species and lucerne in 1996 (Figure 3b) and 1997 (Figure 3c), the water storage capacity for the whole profile in May 1997 was significantly ($P < 0.05$) greater under woody species (248 mm) than lucerne (213 mm) and cropping (80 mm). Figure 3c also suggests drainage under the crop during the intervening period. The Agricultural Production System Simulator (APSIM) described by McCown et al. (1996) was used to evalu-

ate the long term benefits of a dry soil profile after the woody species were removed by simulating a wheat crop. The simulation assumed effective weed control and no run-off, along with other conditions reported by Mele and Yunusa (2001). The results showed there was no drainage beneath 1.8 m depth for at least 4 years (Figure 4c) after woody species, whereas up to 178 mm of the 774-mm annual rainfall was lost as drainage under continuous annual cropping in 1995. Total drainage over 7 years was 420 mm, which was twice that predicted for soil treated with a phase of woody species. Grain yield was reduced only in the first year after removal of the woody species (Figure 4d), as a result of a water deficit (Figure 4a) and was similar to the experimental results obtained after lucerne by Hirth et al. (2001). These model predictions of water dynamics were consistent with the field data in the same area reported by Ridley et al. (2001), where prevention of deep drainage lasted on average about two years for every year of lucerne.

Macropores influence preferential and lateral flows of water in the soil, and have formed the basis for modelling hydrology of the vadose zone of the soil. Macropore properties (length, tortuosity, diameter, density, distribution and orientation), propagation nodes (large decaying tree roots, fissures between soil horizons and larger mesopores), and antecedent soil moisture were used by Sidle et al. (2000) to develop probability density functions to model preferential flow and water movement in a sub-catchments of ≥ 50 m. Influences of root biopores on run-off and waterlogging include enhanced infiltration and K_{sat} through preferential flow paths. This is important under combined gravitational and capillary potential heads at high intensities of water application (Clothier and Green, 1993), which are experienced in high rainfall environments, particularly where a watertable is present. Enhanced infiltration often results in re-distribution of run-off and lateral flow to vertical flow, thus recharging the soil profile. High porosity may be detrimental to productivity or even exacerbate ground water recharge, where preferential flows prevent full recharge of the soil matrix as observed by Sollins and Radulovich (1988) for exceptionally large pores (250 mm wide and about 0.8 m in length).

Root growth

Roots are re-directed from the vertical towards lateral growth on encountering a compacted soil layer, thereby limiting their ability to explore the soil profile

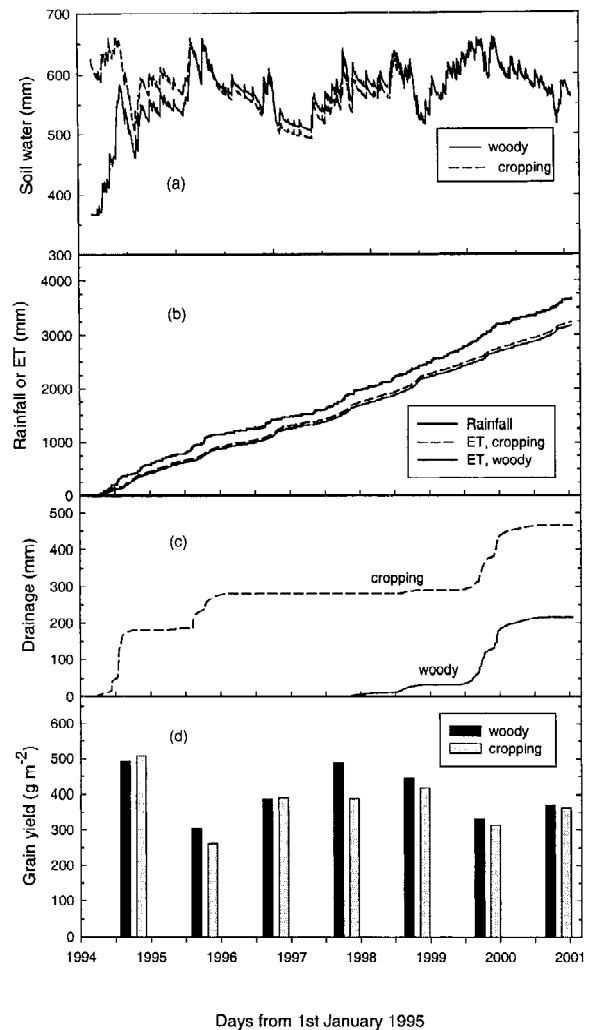


Figure 4. Simulated response of spring wheat to soil conditions under continuous annual cropping or following a 6-year phase of native woody perennials in south-eastern Australia: (a) water stored in the 1.8 m soil profile, (b) cumulative rainfall or evapotranspiration (ET), (c) cumulative drainage, and (d) grain yields.

(Dexter, 1986). Biopores provide minimum resistance to water movement and oxygen diffusion, and therefore provide preferential channels for roots to get through compacted soil layers (Greenland, 1981). Increases in root growth of between 15 and 75% in response to pre-existing biopores have been reported (Comis, 1997; Cornish, 1993; Elkins and van Sickle, 1984; Wang et al., 1986; Yunusa et al., 2001). Response of root growth to biopore creation, however, depends on the availability of soil water. Grecu et al. (1988) found that continuous maize had twice the root length density compared to maize following

Table 5. Responses by root variables to modification of soil physical characteristics

Root response	Crop	Preceding crop or treatment	Value of response	θ (%)	Source
Total root length density (km m^{-3}) in the top 0.55 m profile for no-till soil ^a	Maize	Maize	10.80	na	Hulugalle and Lal (1986)
		Pigeon pea	17.52	na	
Total root length density (km m^{-3}) in 1.05 m profile at tasselling	Maize	Maize	16.10	42	Greco et al. (1988)
		Lucerne + fescue	8.64	34	
Total root length density (km m^{-3}) in 1.2 m profile at tasselling	Soybean	Soybean	3.34	39	
		Lucerne + fescue	4.96	39	
Total root length density (km m^{-3}) in the top 0.8 m soil depth at grain filling	Wheat ^b	Annual crops	0.69	24	Yunusa et al. (2001)
		Woody species	1.47	25	
Root dry matter (g m^{-2}) in the 1.5 m profile at anthesis	Wheat ^b	Annual crops	117.9	25	
		Woody species	168.6	26	
Dry matter (g m^{-2}) in subsoil (0.45–0.65 m depth) at early pod growth	Canola	Clover	33.8	14	Ward et al. (2002)
		Lucerne	15.0	12	

^a θ not given, but most likely to be similar for the two treatments.

^bCrop infested by weeds of annual ryegrass (*Lolium rigidum* Gaudin).
na data not reported.

lucerne–fescue swards, in which the soil was 8% drier (Table 5). Whereas, where the soil water content was similar ($\theta=39\%$) in the continuous soybean comparison, increased root length density of soybean could be associated with biopore formation that preceded the mixed sward. Table 5 also showed that the root mass of canola was 50% less after lucerne compared to that after clover, where the band of subsoil was marginally drier after lucerne (Ward et al., 2002). Once the limiting soil water is overcome, significant enhancement of root growth may result following the priming phase (Hulugalle and Lal, 1986; Yunusa et al., 2001). Also, root growth may not benefit if the continuity of pores is broken by tillage making it difficult for roots to ‘sense’ or find the pores. Such events may trigger responses in roots similar to those associated with adverse subsoil conditions of low water availability and/or aeration, high strength, and chemical toxicity, and can also result in physiological responses such as stomatal closure (Stirzaker et al., 1996).

Root water uptake

Water extraction of wheat due to the heterogeneity in biopore distribution was simulated using biopore density and diameter, and crack spacing and layering of the soil profile (Jakobsen et al., 1989; Jakobsen and Dexter, 1988). The model predicted greater water-

use at depth as root penetration increased with the volume fraction of pores from 0.1% to an optimum at 1.5–2.0% as given by:

$$D = 0.81 \cdot d[\ln(1.0 - P_d)]/H, \quad (2)$$

where the distance D to a biopore is dependent on the diameter of the biopores (d) and fraction of the soil volume (H) occupied by biopores; in Equation (2) P_d is the fraction of roots meeting a biopore within the distance D . Using this scheme, both transpiration and grain yield for wheat were predicted to decline once pore diameter exceeded 0.8 mm or at high numbers ($>1000 \text{ m}^{-2}$) of large pores. This is because of rapid use of water during early growth in environments with limited rainfall. Equation (2) was used to show that transpiration was optimum at approximately 1000 pores m^{-2} for soils with high strength and low plant-available water holding capacity, but was at least 3000 pores m^{-2} for soils with high water-holding capacity (Jakobsen and Dexter, 1988).

Passioura (1991) developed a single-root model to demonstrate the importance of biopores in enhancing water uptake by roots. The single root model was adopted to describe the function of roots confined in the biopores, whereby the rate of water uptake is determined by the density of biopores. Passioura (2000) determined the time constant (T , days) for extracting a given volume of water from the soil to be a quarter of

Table 6. Time constant for water extraction (T) estimated with the model of Passioura (1991) for wheat in response to number of biopores created by treatments prior to sowing

Treatment	Soil depth (m)	Biopores (m ⁻²)	T (days) ^a	Source
Conventional tillage, stubble burnt	0.1	96	3.32	Haines and Uren (1990)
Direct drilled, stubble burnt	0.1	81	3.93	
Direct drilled, stubble retained	0.1	123	2.59	
6-year phase of woody species	0.5	161	1.98	Yunusa et al. (2002b)
Continuous annual crop rotations	0.5	97	3.28	
Permanent ryegrass pasture	0.5	60	5.31	
Continuous wheat cropping	0.15	235	1.35	Cresswell and Kirkegaard (1995)
Canola–wheat rotation	0.15	196	1.62	

^aBased on the arbitrary assumption of biopore length proportional to density where each day represents a proportional rate of extraction $1/e$ in the depletion function after Passioura (1991).

the average distance (cm) between the biopores. The T value declines as a function of the reciprocal of the natural logarithm ($1/e$ or 0.37), and is calculated from a putative cylinder of soil around the biopore, B :

$$T = 2B^2/\delta = 1/[\pi L], \quad (3)$$

where $B = (\pi L)^{1/2}$, L is the length of pore occupied by roots per unit volume of soil (cm/cm³) and the diffusivity of soil water (δ) is approximately constant at 0.02 m² day⁻¹ (Passioura, 1991). The potential application of Equation (3) is given by the data in Table 6 comparing water uptake from soils having variable numbers of biopores greater than 2 mm diameter, assuming a fully occupied pore space and uniformly distributed pore length. The rate of water extraction increased with biopore numbers irrespective of soil depths, such that an increase of 50% in biopore numbers reduced the time needed to extract water by 34% in the study of Haines and Uren (Table 6). Where the roots are clumped due to biopores being much larger than the diameter of roots, the time constant could lengthen to weeks, because heterogeneous distribution of roots in biopores may cause interfacial resistances to water-use, compared to where roots are uniformly distributed (Passioura, 1991).

Nutritional and biological responses

Nutritional and biological responses due to increased macroporosity should be considered in terms of the size and quality of organic matter pools and the associated nutrients, which determines the ability of following crops to utilise these resources. Absence of

tillage operations under perennial primer-plants conserves soil organic matter, although its accumulation is dependent on the primary productivity of plant species and the climate (Noble and Randall, 1998). Biopores modify spatial variability in the distribution of organic matter and nutrients in the soil, to create so-called *hot spots*, where roots tend to preferentially concentrate (Passioura, 2000). A short phase of primer-plants would contribute to the soil nutrient pool through the deposition of root matter, which accounts for 20–25% of total biomass in woody species (Armson, 1977) and contributes as much carbon as the litter-fall from the tops (Nadelhoffer et al., 1985). Layers of remnant cell-walls bordering root biopores, are richer in carbon, nitrogen, phosphorus, iron and manganese than the bulk soil (Kirby et al., 2000), and so support diverse assemblages of bacteria that are capable of using a wide range of substrates (Pierret et al., 1999; Stewart et al., 1999). Biopores could minimise nutrient loss through run-off by providing channels for dissolved or particulate forms of nutrients to move more rapidly into the soil (White, 1985; Sollins and Rudulovich, 1988). They also modify soil chemistry to enhance root function, such as the buffering of soil pH by tree biopores that resulted in a 30% increase in the yield of maize (Hulugalle and Lal, 1986).

A short phase (3–6 years) of perennial species as primer-plants could enhance nutrient conservation through improved cycling of elemental reserves, either inherently resident in, or leached into, the lower depths of the soil profile, and subsequent return to the top layers through litter. Peter and Lehmann (2000) showed that root systems of *Acacia saligna* captured sub-

Table 7. Distribution of selected exchangeable cations and total CEC, and incidence of *Rhizoctonia* spp. observed in the soil profile under annual crop rotations or after six years of native woody species^a

Soil horizon	Ca ²⁺ (cmol kg ⁻¹)		K ⁺ (cmol kg ⁻¹)		CEC (cmol kg ⁻¹)		Rhizoctonia (DNA/g)	
	Annual	Woody	Annual	Woody	Annual	Woody	Annual	Woody
A-Horizon	2.99	2.81	0.37	0.40	4.69	4.19	0.06	0.32*
B-Horizon	6.27	4.74*	0.45	0.41	17.41	14.13*	na	na

^aFrom Mele et al. (2003); *pairs of means in the row are significantly different at $P < 0.05$.

stantial amounts of mobile nutrients such as NO₃⁻, K⁺, Ca²⁺, and Fe²⁺ from lower depths of the soil profile and concentrated them in the foliage. A limited analysis of nutrient distribution in south-eastern Australia, however, suggested that although woody species might extract greater amounts of cations from the subsoil than annual crops, the ions were not returned to the topsoil by woody species (Table 7). This is probably because plants growing on these soils of low nutrient reserves (Naidu and Rengasamy, 1993) have evolved to return limited amounts of scarce elements through litter-fall (A. Noble, pers. comm.). This study also showed that a short phase of primer-plants could provide a disease-break to minimise infestation by significantly reducing the potential for infection of the following crop by *Rhizoctonia* spp. (Table 7). Passioura (2000) has discussed other plant root-microbial associations that are either beneficial or detrimental to crops.

Potential limitations to application of primer-plants

Priming of physical properties may not always produce desirable outcomes in terms of yield and hydrological benefits. As noted earlier, Shipitalo et al. (2000) showed that plant-available water may be reduced by excessive preferential flows in the presence of large pores, with the potential to exacerbate recharge to shallow water-tables. Preferential flow may also leach out nutrients and organic matter beyond the reach of plant roots or from the root zone completely (Addiscot and Thomas, 2000). Excessive drying of soil profiles may cause transient compaction of the soil (Yunusa et al., 2002b), while a build-up of a seed-bank for weeds during the priming phase, could severely penalise following crops. Clumping of roots inside biopores may inhibit lateral exploration of the soil and utilisation of soil resources. Stewart et al.

(1999) found that 80% of the roots of a grassy species were located either directly in the pore space or in the *macropore-sheath*, more than would be expected if the roots were randomly distributed. Pierret et al. (1999) associated such preferential root growth with high nutritive status of the sheath, which could increase the soil resistance component of soil-root water flow by up to 200% compared to where roots were more uniformly distributed (Tardieu et al., 1992). Clumping of roots occurs where fine rooted species grow on soils with large pores, such that the air-gaps inhibit the roots' ability to generate sufficient pressure required to break through the biopore wall and the surrounding dense soil (Stirzaker et al., 1996). This pattern of root growth may limit the ability of the plant to exploit soil resources, such as water, resulting in superficial drought. The inability of wheat roots to break out from biopores probably explained the results reported by Lorimer and Douglas (2001) where a crop on soil previously under forest soil was water-stressed, despite water content of the soil matrix being close to field capacity. Furthermore, biopores and the surrounding sheath may provide habitat for pathogenic microbes, nematodes, bacteria and protozoa, and fungi to proliferate throughout the whole profile (Young and Ritz, 2000). Cresswell and Kirkegaard (1995) have also discussed some of these potential limitations.

Desirable characteristics for an effective primer-plant

In addition to adaptation to climate, an ideal primer-plant needs to be able to tolerate hostile physical and chemical limitations in the subsoil. There are several morphological and physiological characteristics that may confer some degree of tolerance to hostile soil environments.

Adaptation to hostile soil chemistry and waterlogging

Plants adapt to hostile soil chemistry either by complexing the toxic ions, e.g., oxidation of Al^{3+} by gamagrass (Clark et al., 1998), and/or buffering (Noble et al., 1999). Effective primer-plants should be tolerant of hypoxic or anaerobic conditions resulting from restricted oxygen diffusion in dense and/or waterlogged soils. Clark et al. (1998) associated the success of gamagrass in growing through dense subsoil with the roots having aerenchyma tissues that supplied oxygen to the growing root tips thereby enabled them to penetrate wet dense layers. They also found that these roots were tolerant to toxic levels of acidity and Al^{3+} concentrations. Other plants, such as *Acacia confusa* Merrill, maintain physiological processes in a low oxygen environment by slowing down the rate at which they accumulate abscisic acid, and by maintaining a high turgor, unlike less-adapted species such as *Litsea glutinosa* (Liang et al., 1999).

Thick roots

Thick roots generally confer the ability to generate high pressures needed to penetrate dense soils. Field and laboratory studies showed that species with thick roots penetrated dense soils more easily than thin-rooted monocotyledons (Materechera et al., 1991, 1993). This was consistent with the study of Elkins and Van Sickle (1984) in which lines of bahia grass with large roots grew more readily through plough pans than the lines with thin roots. Large roots of gamagrass have been partly credited with the plant's ability to penetrate dense soil (Comis, 1997). Large pores created by thick roots are known to last longer and withstand disturbance due to tillage to a greater extent, than fine pores (Blackwell et al., 1990).

Ease of establishment, perenniality and rapid growth rates

Ability to easily germinate from seed will reduce the costs for the establishment of primer-plants. Perennials have a greater opportunity to punch through impermeable soil layers, by exploiting windows of opportunity that favour root growth such as fissures created by repeated wet-dry cycles (Pillai and McGarry, 1999), compared to the limited growth duration of annuals (Creswell and Kirkegaard, 1995; Elkins and Van Sickle, 1984). For instance, Cornish (1993) reported that root growth by ryegrass seedlings was

significantly increased on soil that previously supported a native perennial grass (*Bothriochloa macra*) compared to soil formerly under annual grasses. Rapid growth rates that result in proliferation of roots in the soil should help to ameliorate the soil within a reasonably short period of time. This is important to allow resumption or commencement of the main economic cropping phase. At the end of soil priming, the plant should lend itself to cheap and easy means of removal that would cause minimal disruption of the soil, e.g., there should be no need for intervention with heavy machinery.

Rapid decomposition of remnant roots

Rapid decomposition of roots is a requirement for quick formation of *biopores*. Rates of decomposition are determined by the C:N ratio, composition of sugars, microbial flora population and thickness of the root. The closer the C:N ratio is to the optimum of 25 (White, 1997), the greater microbial activity and the sooner the formation of biopores. Thick roots of woody species such as *Quercus* spp. and *Pinus* spp. tend to have higher C:N ratios of 70–106 (Usman et al., 2000) and concentrations of insoluble carbohydrates as in *Acacia* spp. (Peter and Lehmann, 2000), than the thin roots of herbaceous species with C:N of 70 (J. Baldock and I. Yunusa, unpublished data).

From the foregoing, therefore, an ideal primer-plant should be a perennial that germinates easily from seeds, grows rapidly, and has thick roots that are adapted to low oxygen environments, but which has some components that decompose quickly once dead. Possession of thick roots or a perennial life cycle on its own is not enough to produce the desired modification of structurally difficult soils. Ilyas et al. (1993) found that the roots of sesbania did not make significant inroads into dense subsoil in the first year of planting, despite being healthy, thick and profusely branched. Gamagrass, however, seems an ideal primer-plant because of its thick aerenchymatous roots, which grow with relative ease through restrictive layers to depths of more than 2.3 m, but die off within 2 years to provide channels for following roots (Clark et al., 1998). In Australia, the challenge is to identify native plant species that have characteristics that are similar to those of gamagrass. The poor results (Table 4) achieved in ameliorating soil impermeability by Creswell and Kirkegaard (1995) were probably because canola was an annual and had thin roots compared to the woody perennials used by Yunusa et al. (2002b).

Table 8. Potential Australian native and exotic primer-plant species with desirable rooting and water-use activity characteristics

Botanical name	Ecology and characteristics	Economic use and management during primer phase
Grasses and herbs		
<i>Phalaris aquatica</i>	Perennial with short rhizomes, tussock forming, drought tolerant, ~2 m tall. Grows best on heavy soils with clay within 0.3 m of the surface.	Pasture in medium rainfall zones of SE Australia (can have toxic alkaloids). Seedlings delicate, difficult to establish.
<i>Microleuca stipoides</i>	Perennial, with short rhizomes and narrow grain (C4). In habits damp grasslands, woodland and forest. Grows 0.1–1.0 m tall.	Little known but used as a low input native pasture, can survive cultivation.
<i>Themeda triandra</i>	Perennial, tufted (C4). Dominant in temperate Australia on most soil types; optimum temp 25–35 °C. Grows 0.3–2.5 m tall.	Low input pasture with younger growth palatable to grazing stock.
<i>Lophopyrum ponticum</i>	Tussock forming, summer active. Grows well in poorly drained, moist soils and is tolerant of salt. Grows to 2 m tall.	Pasture for problem sites. Germinates well from seeds, but plants develop slowly. Needs light grazing in first year.
<i>Austrodonthonia semiannularis</i>	Perennial, tussock forming with grain to 1.5 mm long. Inhabits sea-level to foothills of SE Australia on soils ranging from sands to clays in drier areas	Low input pasture.
<i>Mellilotus</i> spp.	Warm season biennial legume; massive roots that disrupt heavy clay; up to 2 m tall.	Pasture legume. Some varieties high in cumin affect stock health.
<i>Bothriochloa macra</i>	Warm season perennial, prostrate, tufted, leaves of red/purple colour. Widespread drought and frost tolerant. Grows from 0.3–8.0 m tall.	Native pasture species that responds to fertiliser and grazing. Stems of low feed value but leavers are palatable when green.
Shrubs		
<i>Allocasuarina</i> spp.	N fixing legume. Drought tolerant, some species salt tolerant.	Bioenergy source, wood chips for pulp, posts, and possible greenhouse abatement. Management — establishment by seeds or seedlings with rip-line and weed control. Rangeland forage and shelter plants as windbreak for grazing livestock and erosion control. Pollen for apiary industry, fodder and cut flowers. Brush fences.
<i>Melaleuca</i> spp.	Paper barks. Widespread, drought tolerant and some species salt tolerant.	
<i>Banksia marginata</i>	Wide spread; prefers heaths, scrubby forests; redistributes P and water in the soil profile.	
<i>Artriplex nummularia</i>	Dense silver-grey coloured shrub. Grows on heavy soils.	
Trees		
<i>Eucalyptus</i> spp.	Widespread drought and waterlogging tolerant species grow on sandy to heavy clay soils (e.g., <i>E. microcarpa</i>).	Bioenergy source, wood chips for pulp, posts, possible greenhouse abatement. Insect resistant with bark used for extracting tannins. Shade, shelter for livestock and erosion control. Uses include, windbreak fence posts, firewood and apiary industry.
<i>Acacia</i> spp.	About 1150 spp prostrate shrubs to large forest trees; adapted to arid and semi-arid zones and most soil types. N fixing legumes; some noted for tolerance to salinity (<i>A. stenophylla</i>), frost and heavy textured soils.	

Although lucerne has proven ability to dewater the clayey soils beyond 3 m depth (Dunin et al., 2001; Hirth et al., 2001; Ridley et al., 2001), it seems to have limited ability to create effective long-lasting biopores. For instance, Blackwell et al. (1990) observed that lucerne created large biopores on the red-brown earth (Natrixeralf), some as large as 7.5 mm wide, only when the soil was deep ripped to 0.8 m and treated with 10 t ha⁻¹ of gypsum. A similar response was reported by Ilyas et al. (1993) who found that lucerne roots grew beyond the normal depth of 0.8 m only when the soil was treated with gypsum at 2.5 g m⁻². Grecu et al. (1988) also reported that unlike maize (*Zea mays*) and soybean (*Glycine max* L.), 2 years of mixed swards of lucerne and fescue (*Festuca arundinaceae*) confined their roots to the topsoil and did not penetrate the claypan in untreated soil. Even in situations where deep rooting has been observed for lucerne, as suggested by relatively deep (>3.0 m) water extraction (Ridley et al., 2001), its ability to increase macroporosity of dense soils is not certain. Lucerne is also restricted to zones where annual rainfall is between 300 and 600 mm, since below this range it is constrained by limited supply of water, and above the range by acidity (Whitfield, 1998).

There is the need in Australia to explore the potential of perennials, especially of native species for their capacity as primer-plants. Some of the potential candidates given in Table 8, possess several major physiological functions and adaptations of ideal primer-plants. The perception of their limited economic uses may be overcome with new opportunities in paper, bio-fuels, greenhouse gas abatement and biodiversity. Growing them in short phases will minimise soil compaction due to excessive drying and disruption of soil structure during removal. The ability of acacia to open up difficult soils was demonstrated by the work of Fakuoka (1985) who observed that 'five or six years after planting acacias, . . . what had been hard, lean soil. . . had become soft and porous.' It may be necessary, however, to grow defined mixtures of these native species in the priming phase in order to mimic pre-clearing vegetation for optimising both hydrological control and soil improvement objectives. Hobbs and Cramer (2003) identified heterogeneity of vegetation structure at stand and landscape levels as necessary for providing the diverse above- and below-ground diversity that confers resilience on woodland systems. As such, on a duplex redbrown earth in the 400–600-mm rainfall zone, for example, a mixture of species involving *P. aquatica*, *L. ponticum*, *A. num-*

mularia, and acacias will form an effective primer community across landscapes with hydrological and soil degradation problems. Mappin et al. (2003) listed key physiological characteristics, including rooting patterns, for several native species occupying a range of storeys, which could be used to identify potential primers for achieving high rates of water-use and amelioration of dense subsoils.

Knowledge-gaps in implementing the primer-plant concept

In this review we have dealt with the general principles of the primer-plant concept and the likely impact on soil structural characteristics and hydrological processes. Implementation of this concept, especially using Australian native perennials, requires further understanding of several key plant and soil processes and their interactions including:

1. A better understanding of the growth and function of the roots of native species. Although deep rooting is considered a characteristic of most of the woody perennials, it is not certain how the plants adapt to, or tolerate, the huge spatial and temporal variability in soil-water availability and other adverse chemical conditions. This understanding is essential to allow informed choice of candidate primer species appropriate for a given soil and prevailing environmental conditions.
2. Management of woody and herbaceous species for enhanced realisation of priming benefits. Understanding is still limited in terms of agronomy of these species and efficient harvesting regimes and techniques to minimise disturbance and loss of the topsoil. Although Yunusa et al. (2002b) found significant subsoil amelioration after 6 years of woody species, it was not certain whether a similar outcome could be achieved within a shorter time-frame or using higher planting density and/or varied fertility management.
3. Understanding the character of the increased porosity, in terms of interactions between pore-size distribution and density on soil hydrology and on growth and function of roots of the following crops. The complexity of these interactions and their influences made them difficult to evaluate, and models are often employed for this purpose. The modelling work of Dexter and colleagues, as reviewed and discussed above, has provided some insight into the effect of interaction between pore-

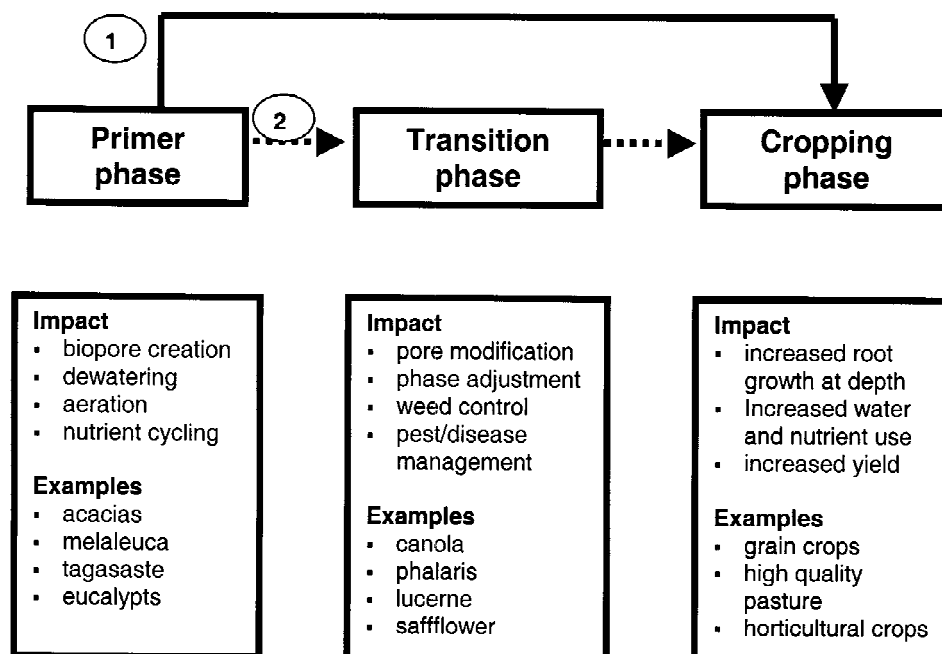


Figure 5. Pathways for implementing the primer-plant concept. In pathway 1 transition phase is not necessary and the cropping phase follows immediately after the soil priming phase. In pathway 2, it may be necessary to either adjust the effects of soil priming or cropping sequence before the main economic crop.

size distribution and density, on the one hand, and root growth and function, on the other. However, these predictions need field validation before they have any practical use.

- Longevity of the priming effects needs to be evaluated, especially given that resumption of annual cropping with associated tillage operations may easily disrupt continuity of pores. This assessment should take into account pore size distribution and length.
- The choice of crop after soil priming may significantly influence the economic viability of this concept. Improvements in the yields of the economic crop following priming should be large enough to more than make up for the loss in production while the soil was being primed. It is therefore important to choose carefully the crop(s) to grow after priming. For instance, as we discussed above, thin fibrous roots of cereals are unlikely to benefit from large biopores created by woody species, as channels for new growth. Crops with high growth rates may be needed to suppress aggressive weed infestation after a build-up of weed seed-banks during the soil-priming phase (Mele and Yunusa, 2001). It may also be

necessary to introduce primer-plants in staggered phases including a transition phase where necessary (Figure 5). For instance, where there are pest and/or disease control issues, a transition phase may be needed to increase options for weed control using herbicide-tolerant crops or those that are poor hosts to root diseases such as take-all of cereals. Also, further modification in soil conditions implemented through a succession of crops with decreasing root thickness may avoid clumping of roots in large biopores leading cropping phase, ie pathway 1 (Figure 5).

Concluding remarks

In this review we have concentrated mostly on ameliorating physical subsoil constraints, which are only a small part of solutions for attaining hydrological balance and profitable farming enterprises. The evidence presented in this review suggests that a short phase of native (and exotic) perennials can be incorporated as primer-plants into cropping systems to improve soil conditions by creating biopores in addition to providing hydrological control through minimising recharge to the water table. We presented evidence on

how a succession of species can be manipulated over time and space to achieve soil structural and hydrological outcomes for productivity gains and sustainable land management. Growing the primer-plants in short-phases minimises problems of soil compaction experienced under tree-dominated vegetation in dry environments, while also minimising length of time the land is removed from economic cropping. Several desirable characteristics for effective primer-plants were presented in this review, including roots that can withstand adverse soil chemistry, thick enough to punch through dense subsoil and decompose rapidly after death. Several knowledge-gaps were discussed, amongst which is the current limited understanding of, and skills in, the agronomy and management of the Australian native woody species. These knowledge-gaps need to be addressed before a wide adoption of the primer-plant concept can be achieved, and identifying new economic uses for the woody species will also facilitate this process. To combat dryland salinity and improve soil properties, a mixture of Australian native woody species can most closely mimic the cleared native vegetation communities, which were complex ecosystems. It should be noted that we are yet to gain a complete understanding of how the historical changes in vegetation are linked to soil structural degradation, and its consequent effects on soil water dynamics.

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